Practical 01

Introduction to Python for Machine Learning

1. Instructions

1. Complete all exercises in **Google Colab**.
2. Create a new notebook named:

   `AML_Practical_01_RegistrationNumber.ipynb`

3. Write Python code for each question in a separate code cell.
4. Execute the code and verify the output.
5. Use meaningful variable names and appropriate comments.
6. Do not hardcode outputs where calculations or programming logic are required.
7. Include suitable Markdown headings to identify each question.
8. Ensure all code cells run successfully in Colab. Save your notebook to Google Drive, and download the completed `.ipynb` file using **File** → **Download** → **Download .ipynb**. Submit the notebook to the VLE.

2. Python Programming Exercises

Complete the following exercises using Python in Google Colab.

Question 1: Variables and Arithmetic Operations

Consider the following variables:

```
student_name = "Amal"
assignment_marks = 75
exam_marks = 85
```

Write a Python program to:

a) Display the student's name.

b) Calculate the total marks.

c) Calculate the average marks.

d) Display the data type of each variable.

Expected Results:

```
Student Name: Amal
Total Marks: 160
Average Marks: 80.0
```

Question 2: Type Conversion

Consider:

```
num1 = "25"
num2 = 15
```

Write a Python program to:

a) Display the data type of `num1`.

b) Convert `num1` into an integer.

c) Calculate the sum of `num1` and `num2`.

d) Display the result.

Question 3: Predict the Output

Without executing the following code, predict its output.

```
x = 10
y = 3

print(x / y)
print(x // y)
print(x % y)
print(x ** y)
```

Explain the difference between the operators:

`/`, `//`, `%`, `**`

Question 4: List Operations

Consider the following list:

```
marks = [78, 85, 62, 90, 74]
```

Write Python statements to:

a) Display the first element.

b) Display the last element.

c) Change the third element to 70.

d) Add 88 to the list.

e) Calculate the sum of all marks.

f) Find the highest and lowest marks.

Question 5: List Indexing and Slicing

Consider:

```
numbers = [10, 20, 30, 40, 50, 60, 70, 80]
```

Write Python statements to:

a) Extract the first three elements.

b) Extract the last three elements.

c) Extract elements from index 2 to index 5 (inclusive).

d) Extract every second element.

e) Reverse the list using slicing.

Question 6: Tuples

Consider:

```
coordinates = (10, 20, 30, 40)
```

Write Python statements to:

a) Display the second element.

b) Display the number of elements.

c) Convert the tuple into a list.

d) Change the first element of the converted list to 50.

e) Convert the modified list back into a tuple.

Question 7: Dictionaries

Consider the following dictionary:

```
student = {
    "name": "Nimal",
    "age": 22,
    "course": "Computer Science",
    "marks": 82
}
```

Write Python statements to:

a) Display the student's name.

b) Update the marks to 90.

c) Add a new key called `"grade"` with the value `"A"`.

d) Display all keys.

e) Display all values.

f) Check whether the key `"email"` exists in the dictionary.

Question 8: Student Grade Calculator

Write a Python program that accepts a student's marks and displays the corresponding grade. Use the following grading scheme:

Marks Range	Grade
75–100	A
65–74	B
55–64	C
40–54	D
Below 40	F

Your program should also handle invalid marks (less than 0 or greater than 100).
Example:

```
Enter marks: 78
Grade: A
```

Question 9: Even or Odd

Write a Python program that accepts an integer and determines whether it is even or odd.
Example:

```
Enter a number: 15
Odd
```

Question 10: Conditional Expressions

Predict the output of the following program.

```
x = 15
y = 20

if x > 10 and y < 25:
    print("Condition 1")

if x > 20 or y == 20:
    print("Condition 2")

if not (x == 15):
    print("Condition 3")
else:
    print("Condition 4")
```

Explain why each condition is evaluated as true or false.

Question 11: For Loop

Consider:

```
numbers = [5, 10, 15, 20, 25]
```

Write a Python program to:

a) Print every element using a `for` loop.

b) Calculate the sum of all elements without using the built-in `sum()` function.

c) Count the number of elements greater than 12.

Question 12: While Loop

Write a Python program using a `while` loop to display the numbers from 1 to 10.
Modify the program to calculate the sum of the numbers from 1 to 10.
Expected Result:

```
Sum: 55
```

Question 13: Filtering Values Using Loops

Consider:

```
marks = [45, 78, 32, 90, 56, 28, 65]
```

Write a Python program to:

a) Create a new list containing only marks greater than or equal to 50.

b) Count the number of marks below 50.

c) Calculate the average of the passing marks.

Expected Output:

```
Passing Marks: [78, 90, 56, 65]
Number of Failures: 3
Average Passing Marks: 72.25
```

Question 14: Simple Function

Write a Python function named `calculate_area()` that:

a) Accepts the length and width of a rectangle.

b) Calculates its area.

c) Returns the calculated area.

Call the function with length 10 and width 5.
Expected Output:

```
Area: 50
```

Question 15: Calculating the Mean

Write a Python function named `calculate_mean()` that accepts a list of numbers and returns their arithmetic mean.
Do not use NumPy or the `statistics` library.
Test the function using:

```
numbers = [10, 20, 30, 40, 50]
```

Expected Output:

```
Mean: 30.0
```

Question 16: Functions and Conditional Statements

Write a Python function called `check_pass()`.
The function should:

a) Accept a student's marks as a parameter.

b) Return `"Pass"` if marks are greater than or equal to 50.

c) Return `"Fail"` otherwise.

Use the function to check:

```
marks = [75, 40, 65, 30, 90]
```

Expected Output:

```
75 - Pass
40 - Fail
65 - Pass
30 - Fail
90 - Pass
```

Question 17: Math Library

Import Python's built-in `math` library.
Write a program to:

a) Calculate the square root of 144.

b) Calculate the factorial of 5.

c) Display the value of π.

d) Calculate the ceiling of 4.3.

e) Calculate the floor of 4.8.

Question 18: Random Library

Import Python's built-in `random` library.
Write a Python program to:

a) Generate a random integer between 1 and 100 (inclusive).

b) Randomly select one element from the following list:

```
colors = ["Red", "Blue", "Green", "Yellow"]
```

c) Shuffle the elements of the list.

Note: The output may vary each time the program is executed.

Question 19: NumPy Array Operations

Create the following NumPy array:

```
[10, 20, 30, 40, 50]
```

Perform the following operations:

a) Display the array.

b) Multiply every element by 2.

c) Calculate the mean.

d) Calculate the maximum and minimum.

e) Display the number of dimensions.

f) Display the shape.

Question 20: Student Performance Analysis

Consider the following dataset:

```
students = [
    {"name": "Amal", "marks": 75},
    {"name": "Nimal", "marks": 45},
    {"name": "Kamal", "marks": 85},
    {"name": "Sunil", "marks": 35},
    {"name": "Saman", "marks": 90}
]
```

Write a Python program to:

1. Extract all marks into a list.
2. Calculate the average marks.
3. Find the highest and lowest marks.
4. Identify students who passed (marks greater than or equal to 50).
5. Count the number of students who failed.
6. Create a Pandas DataFrame using the given data.
7. Display a bar chart showing student marks.

End of Practical 01